

PRODUCT REQUIREMENTS DOCUMENT

AI-Powered Crime Response Intelligence Platform

Hackathon MVP — Draft v1.0

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Status: Draft for development kickoff

1. Overview & Problem Statement

Crime response speed is limited less by information availability than by how quickly fragmented information — citizen reports, CCTV detections, and police records — can be verified, prioritized, and turned into a dispatch decision. Today this fragmentation forces manual verification and coordination before any action is taken, slowing down the critical early stage of a response.

This platform is an AI Intelligence Layer, not a replacement for existing reporting channels. It aggregates crowdsourced reports (e.g. Super App Polri), CCTV computer-vision detections, and (offline) Body-Worn Camera footage into one operational view, then supports operators with correlation, severity classification, AI-generated summaries, an interactive hotspot map, and nearest-unit dispatch recommendations across a structured incident lifecycle.

1.1 Goals

- Reduce the time between incident detection and a dispatch decision.
- Give operators one unified, de-duplicated view instead of several disconnected report streams.
- Surface actionable recommendations (severity, nearest unit) without forcing operators to manually cross-reference sources.
- Keep humans firmly in control of execution (radio dispatch) while the system accelerates judgment.

1.2 Non-Goals (for this MVP)

- Replacing Super App Polri or any existing citizen-facing reporting app.
- Automating radio/HT dispatch communication itself.
- Building or training a production-grade computer-vision or BWC model — the hackathon build uses simulated/mocked detections and an LLM-based classifier/summarizer instead of a custom-trained model.

2. User Roles

Role	Description
Operator / Dispatcher	Primary user. Monitors the hotspot map and notification panel, opens incident detail, verifies incidents, selects and assigns nearest personnel, confirms lifecycle transitions.
Supervisor	Secondary user. Views aggregate dashboard/stats, can review historical incidents and response-time performance. Read-heavy role in this MVP.
Field Personnel (BWC wearer)	Not a system UI user in this MVP — represented in the system only through their BWC's live GPS feed and their status field. Communication with them is via HT, outside the platform.

3. Finalized User Flow

This is the single, agreed end-to-end flow the MVP is built against. It intentionally keeps two entry points into the same modal, and separates "browsing personnel" from "computing nearest distance" so the distance calculation only runs when explicitly requested — not on every page load or filter change.

1. Operator views the Heatmap, with a Latest Incidents notification panel above/beside it.
2. Operator opens an incident either by (a) clicking a hotspot point on the map, or (b) clicking an item in the notification panel — both open the same Incident Detail Modal.
3. Modal shows: timeline, category, severity, verification status, AI-generated summary, and linked evidence.
4. Operator clicks "Assign Personnel Terdekat" inside the modal.
5. System navigates to the Personnel List page, carrying the incident_id in context. The list loads unsorted by default (no distance calculation runs yet).
6. Operator chooses one of two paths: (a) click "Filter Terdekat" to run the on-demand nearest-distance calculation and re-sort the list, or (b) use Search by ID/Name if they already know who to call.

7. Operator selects the personnel to assign (busy/assigned personnel are badge-flagged but still selectable).
8. Operator clicks "Assign." The system updates incident status to Assigned and generates a Dispatch Card summarizing the incident for radio read-out.
9. Operator manually communicates with the officer over HT/radio — outside the system.
10. Once the officer acknowledges over radio, the operator manually clicks "Confirm En Route," moving the incident to En Route.
11. The operator is returned to the Heatmap/notification view, where the incident now shows its updated status badge — closing the loop for continuous monitoring.

3.1 Why this flow (design rationale)

- Two entry points (map + notification list) into one modal avoids duplicating detail-view logic and matches how operators naturally scan for incidents — spatially or chronologically.
- Splitting the personnel list from the nearest-distance calculation avoids re-running distance computation on every render or filter interaction; it only runs when the operator explicitly asks for it, or is bypassed entirely via search when the operator already knows their target.
- The Dispatch Card + manual "Confirm En Route" step keeps the HT/radio communication — which the system cannot integrate with — explicit and human-owned, so the platform never implies it dispatched a unit automatically.

4. Functional Requirements

Priority uses MoSCoW (Must / Should / Could / Out of scope for this MVP).

4.1 Module A — Data Ingestion & Integration

ID	Requirement	Description	Priority
FR-A1	Simulated multi-source data feed	Generate periodic mock incidents (crowdsourced report format + CV-detected events) with lat/long, category, timestamp, source tag, for demo purposes since live Super App Polri / CCTV APIs are unavailable.	Must
FR-A2	Crowdsourced report ingestion	Accept structured citizen reports (location, category, free-text description, media) mirroring Super App Polri's report schema.	Must
FR-A3	CV event ingestion	Accept structured detection events from a CCTV computer-vision pipeline (e.g. crowd, altercation, suspicious vehicle) with confidence score, camera ID, and coordinates.	Should
FR-A4	Data normalization	Map all incoming sources into one internal schema (see Section 7) so downstream AI processing is source-agnostic.	Must

4.2 Module B — AI Processing Engine

ID	Requirement	Description	Priority
FR-B1	Deduplication & correlation	Group raw reports/events referencing the same real-world incident using a rule-based match: geo-radius (default 150m) + time window (default 30 min) + category similarity.	Must
FR-B2	Severity classification	Assign a severity level (Low / Medium / High / Critical) to each incident using an LLM-based classifier prompted with incident details.	Must

ID	Requirement	Description	Priority
FR-B3	AI incident summary	Generate a concise narrative summary of an incident from its correlated source reports, citing which reports contributed.	Must
FR-B4	Escalation / trend prediction	Flag areas with rising incident density as potential emerging hotspots.	Could

4.3 Module C — Hotspot Map & Visualization

ID	Requirement	Description	Priority
FR-C1	Interactive hotspot map	Render incidents as a clustered heatmap on an interactive map (pan/zoom).	Must
FR-C2	Latest-incidents notification panel	Persistent panel above/beside the map listing newest incidents in real time (or short-interval polling), independent entry point from the map itself.	Must
FR-C3	Filter by category / severity / time	Let the operator narrow the map view.	Should
FR-C4	Live unit position overlay	Show active patrol units on the map, position sourced from BWC GPS feed (see FR-E1).	Should
FR-C5	Map/notification click → detail modal	Clicking a hotspot point OR a notification-list item opens the same Incident Detail Modal.	Must

4.4 Module D — Incident Detail & Verification

ID	Requirement	Description	Priority
FR-D1	Incident detail modal	Displays timeline, category, verification status, severity, AI summary, and linked evidence for the selected incident.	Must
FR-D2	Evidence viewer	Show photos/video/text snippets from correlated source reports within the modal.	Should
FR-D3	Verification status control	Operator can mark incident Unverified / Verified / False report.	Must
FR-D4	Assign Nearest Personnel button	Entry point from the modal into the personnel-assignment flow; carries incident_id forward.	Must

4.5 Module E — Dispatch & Nearest-Personnel Assignment

ID	Requirement	Description	Priority
FR-E1	Live BWC location feed	Ingest periodic GPS pings from each on-duty officer's Body-Worn Camera as the live location source for that unit (distinct from BWC's offline training-data role — see Section 8).	Must
FR-E2	Personnel list page (default unsorted)	On navigating from the modal, show all active personnel with status, last-ping freshness, and current assignment badge — no distance calculation runs automatically.	Must
FR-E3	"Filter Terdekat" (on-demand nearest)	Only when clicked, compute distance from each active unit's last known BWC coordinate to the incident location and re-sort the list. Avoids recalculating on every list render/filter.	Must

ID	Requirement	Description	Priority
FR-E4	Search personnel by ID / name	Lets an operator who already knows who to call jump straight to that person instead of waiting for sorting.	Must
FR-E5	Busy/assigned indicator	Personnel already linked to an active incident show a visual badge; operator may still override and select them.	Should
FR-E6	Select & Assign	Operator selects one (or more, as backup) personnel and confirms assignment; incident status moves Verified → Assigned.	Must
FR-E7	Dispatch card generation	On assignment, generate a short, radio-ready summary card (location, category, severity, incident ID) for the operator to read out over HT.	Must
FR-E8	Manual "Confirm En Route" control	Because HT communication happens outside the system, the operator manually confirms once the officer acknowledges over radio; this transitions status Assigned → En Route.	Must
FR-E9	Auto-return to heatmap after assignment	After assignment is confirmed, the operator is returned to the heatmap/notification view with the incident's updated status badge visible, closing the loop for continuous monitoring.	Should

4.6 Module F — Incident Lifecycle

ID	Requirement	Description	Priority
FR-F1	Incident lifecycle state machine	Enforce the sequence: Reported → Verified → Assigned → En Route → On Scene → Resolved.	Must
FR-F2	Manual status transition	Operator (or field-side proxy) updates status at each stage; no stage can be skipped without an explicit override.	Must
FR-F3	Status change audit log	Every transition is timestamped and stored for response-time analytics.	Should

4.7 Module G — BWC Training Data (documented, not built for demo)

ID	Requirement	Description	Priority
FR-G1	BWC footage as training corpus	Historical BWC recordings are referenced in the data pipeline as offline training/context material for improving classification and summary quality — not a live feature and not built/demoed.	Out of scope

4.8 Module H — Admin, Auth & Overview

ID	Requirement	Description	Priority
FR-H1	Role-based access	Operator vs Supervisor see different views/permissions.	Should
FR-H2	Dashboard overview stats	Active incidents, resolved-today count, average response time, shown on landing.	Should
FR-H3	Authentication	Basic login for demo purposes.	Should

5. Non-Functional Requirements

Category	Requirement
Performance	Distance-ranking calculation ("Filter Terdekat") must return within ~1s for the demo's active-unit pool; it must not run automatically on page load or on every keystroke of the search box.
Performance	Heatmap should reflect a new incident within a short polling interval (target: ≤10s) or via a live update channel.
Scalability	Schema and queries should remain workable at a simulated load of hundreds of concurrent incidents/personnel, even though the hackathon demo dataset will be small.
Reliability	A BWC feed with no ping for longer than a configurable threshold (e.g. 5 minutes) must be marked "stale" rather than silently treated as current — prevents recommending an officer whose position is out of date.
Security	Role-based access control between Operator and Supervisor views; location data treated as sensitive and access-logged.
Privacy	Live BWC location access is scoped strictly to dispatch decision-making, not continuous individual monitoring — this should be a stated policy note, not just an implementation detail.
Auditability	Every incident status transition and assignment action is timestamped and attributable to a user, to support after-action review and response-time metrics.
Usability	The gap between an automated recommendation and a manual, human-executed radio dispatch must be visually explicit in the UI (e.g. the Dispatch Card and the "Confirm En Route" step), so operators never assume the system dispatched a unit by itself.

6. Nearest-Unit Distance Calculation — Technical Note

"Nearest personnel" ranking can be computed two different ways, and the choice affects both accuracy and why the on-demand "Filter Terdekat" button matters:

- Straight-line distance (Haversine formula) — computes distance directly between two GPS coordinates. Extremely cheap; can be run repeatedly with no perceptible lag. Recommended for the hackathon MVP.
- Real routing distance (road-network shortest path, e.g. Dijkstra/A* over a road graph, or an external routing API) — more accurate travel-time estimate, but computationally heavier and typically requires a dedicated routing engine rather than a naive repeated graph search.

The on-demand "Filter Terdekat" button is valid and useful under either approach, since it also improves UX by letting an operator who already knows their target skip the sort entirely — but it becomes a hard performance requirement specifically if a road-network/graph-based method is chosen later. For the MVP, Haversine on the live BWC coordinates is the recommended default.

7. Database Schema

Simplified logical schema — field types and constraints to be finalized during technical design, but table boundaries and relationships below should hold.

Table	Key Fields	Notes
users	id (PK), name, role [operator supervisor], email, password_hash, created_at	System accounts for dashboard access.
personnel	id (PK), name, badge_no, unit_type, bwc_device_id (FK → bwc_devices), current_status [available assigned en_route on_scene offline], updated_at	Patrol officer / unit registry.
bwc_devices	id (PK), personnel_id (FK), last_lat, last_lng, last_ping_at, connection_status [live stale offline]	Live GPS source only. Separate in purpose from bwc_training_clips below — see Section 8.

Table	Key Fields	Notes
reports	id (PK), source [superapp_polri], reporter_ref, category, description, lat, lng, media_url, reported_at	Raw crowdsourced citizen report, pre-correlation.
cv_events	id (PK), camera_id, detected_category, confidence_score, lat, lng, detected_at	Raw CV-detected event, pre-correlation.
incidents	id (PK), category, severity [low med high critical], status [reported verified assigned en_route on_scene resolved], lat, lng, verification_status, created_at, updated_at	Core aggregated entity operators act on.
incident_sources	id (PK), incident_id (FK), source_type [report cv_event], source_id, matched_at	Junction table linking an incident to every raw report/event correlated into it (dedup trail).
ai_summaries	id (PK), incident_id (FK), summary_text, cited_source_ids, model_version, generated_at	AI-generated narrative; keeps a record of which sources it was built from.
evidence	id (PK), incident_id (FK), type [photo video text], url, source_report_id (FK, nullable)	Media surfaced in the Incident Detail Modal.
assignments	id (PK), incident_id (FK), personnel_id (FK), assigned_at, confirmed_en_route_at, on_scene_at, resolved_at, dispatch_card_snapshot	One row per assignment attempt; supports reassignment history.
incident_status_log	id (PK), incident_id (FK), from_status, to_status, changed_by (FK → users), changed_at	Full audit trail for response-time analytics.
bwc_training_clips	id (PK), personnel_id (FK, nullable), recorded_at, storage_ref, labeled_category, used_in_model_version	Historical footage metadata used offline for model development — not queried at runtime.

7.1 Key relationships

- incidents is the core entity; reports and cv_events are raw inputs that get linked to it via incident_sources once the correlation/dedup step (FR-B1) matches them.
- personnel.bwc_device_id points to bwc_devices, which holds only the live GPS feed used for FR-E1–E3 — this is intentionally a separate table from bwc_training_clips, which holds offline historical footage metadata used only during model development (Section 8), never read at request time.
- assignments records each assignment attempt (supports reassignment/history), while incident_status_log is the full state-machine audit trail used for response-time analytics.

8. AI & Data Context

8.1 Data sources

- Crowdsourced citizen reports (Super App Polri format) — simulated for the demo.
- CCTV computer-vision detections (category, confidence, camera location) — simulated for the demo; in production this would come from a live CV pipeline, not built as part of this MVP.
- Body-Worn Camera (BWC), used in two distinct ways that must not be conflated:
 - Live GPS feed — real-time coordinate pings while an officer is on duty, used purely for nearest-unit recommendation (Section 6, FR-E1).
 - Historical footage — offline recordings referenced only during model development to help the AI recognize real-world policing scenarios and improve classification/recommendation quality; not queried or streamed at runtime.

8.2 AI processing pipeline

- Normalization — all raw sources are mapped to one internal incident-candidate schema.

13. Deduplication & correlation — rule-based matching (geo-radius + time window + category similarity) groups raw reports/events that likely describe the same real-world incident. This is deliberately rule-based rather than a trained model for the MVP, to keep it explainable and fast to implement.
14. Severity classification — an LLM is prompted with the correlated incident's details (category, description, number of corroborating reports, keywords) to assign a severity level. This uses prompt engineering against a general-purpose LLM API rather than a custom-trained classifier, appropriate for hackathon timelines.
15. Summary generation — the same LLM is prompted to produce a short narrative summary of the incident from its correlated source reports, and the summary keeps a reference to which source reports it was built from (ai_summaries.cited_source_ids) for operator trust/traceability.

8.3 Role of BWC footage in model development

Historical BWC recordings are proposed as contextual training/fine-tuning material to help the severity-classification and summarization steps better recognize real policing scenarios over time. For this MVP this is a documented future direction only — no BWC footage is ingested, processed, or demoed live. Teams should not build a BWC video-processing pipeline for the hackathon; describing this pathway in the pitch is sufficient.

8.4 Explainability & operator trust

- Every AI-generated severity level and summary should be traceable back to the specific reports/events that produced it (via incident_sources and ai_summaries.cited_source_ids), so an operator can sanity-check the AI's output rather than treat it as a black box.
- The Dispatch Card (FR-E7) is generated from structured incident fields, not free-form LLM output, to keep radio-read-out information accurate and consistent.

9. Success Metrics

Metric	Why it matters
Time from Reported → Verified	Measures how much AI correlation/summary reduces manual verification effort.
Time from Verified → Assigned	Measures how much the nearest-unit recommendation shortens dispatch decision time.
Time from Assigned → On Scene	End-to-end operational response time — the ultimate outcome metric.
% reduction in duplicate reports surfaced to operator	Measures value of the deduplication/correlation step.
Operator task completion rate during demo/user testing	Basic usability signal for the assign-personnel flow.

10. Assumptions & Constraints

- Live Super App Polri and CCTV CV APIs are not accessible during the hackathon; all such data is simulated/mockd but shaped to match realistic schemas so the integration story remains credible.
- BWC live GPS is simulated as a moving "virtual unit" (e.g. periodic randomized movement within a bounded area) rather than real hardware streaming.
- Haversine straight-line distance is the assumed default for nearest-unit ranking; real road-network routing is a noted future upgrade, not an MVP requirement.
- HT/radio communication is and remains fully manual and outside the system for this MVP — no radio integration is attempted.

10.1 Out of scope for this MVP

- Real HT/radio system integration.
- Production-grade, trained computer-vision model for CCTV analysis.
- BWC video ingestion/processing pipeline.

- Native mobile app for field officers (assume web-only for this build).
- SSO / integration with official Polri identity systems.

11. Tech Stack

Layer	Technology
Frontend	Next.js 15 + React 19 + TypeScript
UI	Tailwind CSS + shadcn/ui
State	TanStack Query + Zustand
Forms	React Hook Form + Zod
Maps	Mapbox GL JS + Deck.gl + Turf.js
Charts	Apache ECharts
Backend	Next.js
AI Service	FastAPI
Database	PostgreSQL + PostGIS
ORM	Prisma
Object Detection	YOLOv11
Tracking	ByteTrack
Video Understanding	Video-LLaMA 3
Vision-Language	Qwen2.5-VL
LLM	GPT-5.5 or Qwen3
OCR	PaddleOCR
Storage	Cloudinary
Realtime	Socket.IO
Queue	BullMQ
Reports	pdf-lib or React PDF